Joelvis Miguel Jerez

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EDUCATION

University of Notre Dame | Notre Dame, IN — May 2027
Bachelor of Science in Mechanical Engineering
Relevant Courses: Experimental Robotics, Robotics & Controls, Fluid Mechanics, Gas Turbines & Propulsion
Notre Dame London Global Gateway | London, United Kingdom — Spring 2026

EMPLOYMENT

Barletta Pontoon Boats | Manufacturing Engineering Intern — June 2026 – August 2026
- Audited production CAD drawings for 67 boat models across four product lines and validated base models against manufactured boats, standardizing print formats for the manufacturing floor
- Evaluated handheld and mobile 3D-scanning approaches for per-boat as-built floorplans against accuracy, repeatability, takt-time and scalability requirements, advancing the project to industrial-grade evaluation
- Developed concepts for a universal, self-leveling bracket fixture so any pontoon can run on any line, improving operator safety and supporting future automation; direction approved for prototyping
- Built 3D boat models to support product visualization and digital documentation

Broadway Mercer Associates | Handyman Intern — June 2025 – August 2025
- Diagnosed and repaired plumbing, electrical and carpentry issues in commercial units, building strong attention to detail

TutorND | K–8 Math Tutor — December 2024 – Present
- Teach groups of 2–4 students, building the ability to communicate clearly and break hard concepts into simple steps

RELATED PROJECTS

Gas Turbine Engine Design Studies | Gas Turbines & Propulsion, London — January 2026 – April 2026
- Delivered six team engine-design presentations over the semester, each introducing a new unit of the course
- Wrote Python cycle-analysis code to size a 100 lbf turbofan for Mach 0.8 at 35,000 ft, running trade studies on bypass ratio and fan and compressor pressure ratios (TSFC 0.88 lbm/hr·lbf)
- Built velocity-triangle analysis code and analyzed the tip slice of a helium axial compressor stage (20 kg/s, pressure ratio 1.7), holding relative inlet Mach under the 1.2 limit

Heat Sink Design & CFD Validation | Lab II — November 2025 – December 2025
- Validated a SolidWorks Flow Simulation model against a natural-convection experiment on an aluminum plate-fin heat sink (thin-film heater, thermistor, thermal camera), matching measured base temperature within 3.1 K
- Ran a mesh refinement study from 4,600 to 266,000 cells to choose a model that balanced accuracy and solve time
- Iterated three fin geometries (count, spacing, height, thickness), cutting base temperature 8.1 K (314.5 → 306.4 K) by trading a 25% mass increase for more convective area

AI Maritime Maneuver Indiana Collegiate Challenge (AIMM ICC) | Mechanics — August 2025 – December 2025
- Designed the electronics cooling system and optimized thruster placement for a GPS-guided, low-profile autonomous vessel, coordinating across mechanical, electrical and computing subsystems
- Worked with navigation and control hardware including a Pixhawk flight controller, CAN bus and LoRa/ELRS radio links

Robotic Arm Kinematics & Path Planning | Intro to Robotics — October 2025 – November 2025
- Modeled and validated a 5-DOF robotic arm by measuring link geometry, defining reference frames, and implementing Product of Exponentials forward kinematics in MATLAB
- Developed inverse kinematics and Arduino control code to guide the end-effector through via points, enabling the arm to physically write letters and evaluating discrepancies between theory and hardware

LEADERSHIP AND SERVICE

Society of Latino Engineers and Scientists (SOLES) | Notre Dame, IN
Treasurer — August 2025 – May 2026
- Managed funding requests and reimbursements for professional development events, outreach programs and national conferences such as the SHPE National Convention

Community Outreach Chair — August 2024 – May 2025
- Organized 2–3 community service opportunities per semester for members across Notre Dame and the Michiana area

SKILLS

Certification: Certified SOLIDWORKS Associate (CSWA)
CAD & Simulation: SolidWorks (CAD/CAM), PTC Onshape, Autodesk Fusion 360, SolidWorks Flow Simulation
Programming: MATLAB, Python, C/C++ (Arduino and microcontrollers), Microsoft Office
Fabrication & Lab: 3D printing, CNC milling (CAM), metalworking and machining
Languages: Spanish (fluent) | **Interests:** Cars and car audio, basketball, baseball, dancing, Legos